

OpenBanking/PFM

#	Bài toán cụ thể	Function / journey	Data signal	AI/DS/ML	Output	Được gì	Rủi ro / mất gì
1	Simulated Open Banking account aggregation	Connect bank / sync account	account_id, balance, account type, provider, consent scope	SE + DS pipeline	normalized account data	Học Open API/data integration foundation	Không có API bank thật, phải simulate
2	Consent scope analyzer	User cấp quyền truy cập dữ liệu	consent type, expiry, data scope, provider	Rule + DS	consent summary / risk note	Học privacy/consent mindset	AI/ML nhẹ, nên làm module phụ
3	Consent expiry / refresh reminder	Open banking connection maintenance	consent expiry date, last sync, failed sync count	Rule + DS	refresh reminder	Practical, dễ demo	Không đủ AI/ML nếu đứng riêng
4	Bank data sync failure classifier	Account sync error	error code, provider, retry count, latency, auth status	ML classification	sync failure reason + next action	Học API integration ops	Hơi backend/ops, cần data tốt

5	Transaction normalization across banks	Ingest transactions từ nhiều nguồn	raw description, merchant, amount, date, provider format	NLP/rule + DS	normalized transaction schema	Rất thực tế data integration	Nếu chỉ mapping rule thì AI nhẹ
6	Merchant name normalization	Transaction enrichment	raw merchant text, aliases, location, MCC giả lập	NLP/fuzzy matching/embedding	canonical merchant name	Function nhỏ nhưng rất thực tế	Cần tạo alias dataset tốt
7	Transaction deduplication	Aggregated transaction cleanup	transaction_id, amount, timestamp, merchant, provider	Entity matching/anomaly	duplicate / unique	Hữu ích khi sync nhiều source	AI/ML có thể nhẹ
8	Transaction categorization	Account transaction analysis	description, merchant, amount, MCC, transaction type	ML/NLP classification	category: food/bill/rent/salary	Foundation mạnh cho PFM	Có thể quá dễ nếu đứng riêng

9	Category correction learning	User sửa category	original category, corrected category, merchant pattern	Online learning/ML-lite	improved category rule/model	Có personalization loop	Phức tạp nếu làm online learning thật
10	Recurring bill detection	Detect hóa đơn định kỳ	merchant, amount similarity, date cycle, transaction frequency	DS/time-series/rule	upcoming bill reminder	Function nhỏ, thiết thực	Nếu chỉ heuristic thì AI/ML nhẹ
11	Subscription detection	Detect subscription/auto-pay	recurring card/payment, merchant, amount, cycle	DS/rule + ML-lite	subscription list	Hữu ích với user, dễ demo	Cần synthetic subscription data
12	Bill amount anomaly detection	Hóa đơn tháng này bất thường	bill merchant, historical amount, category baseline	Anomaly detection	bill spike alert	Cụ thể, tốt hơn anomaly chung chung	Cần define “normal” tốt

13	Unusual spending detection	Daily/monthly spending monitoring	amount, category, merchant novelty, user baseline	Anomaly detection	unusual spending alert	DS/ML rõ, practical	Synthetic labels khó
14	Merchant novelty detection	User giao dịch với merchant lạ	merchant first_seen, amount, category, location	Rule + anomaly	new merchant alert	Hữu ích cho PFM/risk nhẹ	Dễ trùng fraud nếu không định vị PFM
15	Large transaction explanation	Giao dịch lớn hơn bình thường	amount vs user baseline, category, merchant	DS + LLM explanation	explanation note	Kết hợp DS + AI tốt	Nếu LLM không grounded thì yếu
16	Monthly spending summary	End-of-month report	category spend, merchant spend, trend vs last month	DS analytics + LLM summary	monthly insight report	Dễ demo, user-facing	Nếu chỉ summary text thì quá nhẹ

17	Spending trend detection	Monthly/weekly trend	category time series, rolling average, seasonality	DS/time-series	trend up/down insight	Học analytics thực tế	Không phải ML mạnh
18	Budget overrun prediction	Trong tháng, dự báo vượt ngân sách	current spend, days left, historical spend pace	Regression/time-series	budget overrun risk	Output cụ thể, có prediction	Cần giả lập budget/user goal
19	Smart budget recommendation	User tạo ngân sách	past spending, income, category distribution	DS/recommendation	suggested budget per category	Hợp personalization	Có thể thành rule nếu đơn giản
20	Income detection	Identify salary/income	incoming transactions, amount stability, employer text	DS/pattern mining	income events	Hữu ích cho credit/PFM	Transaction description mơ hồ
21	Income stability scoring	PFM/affordability	income frequency, variance, missing salary months	DS/time-series	income stability score	Kết nối tốt với credit risk	Cần transaction history đủ dài

22	Cashflow forecasting cá nhân	Dự báo balance ngắn hạn	income schedule, bills, spending pace, balance	Forecasting/regression	7/14/30-day balance forecast	Rất practical, có DS rõ	Forecasting có thể khó nếu ôm sâu
23	Low balance early warning	Trước khi user hết tiền	forecast balance, upcoming bills, income date	DS/forecasting + rule	low balance alert	Output rõ, có ích	Dễ thành rule nếu không forecast
24	Cashflow stress detection	Phát hiện áp lực tài chính	salary delay, bill increase, low balance, overdraft giả lập	Anomaly/ML classification	cashflow stress flag	Có financial health angle	Nhạy cảm, không nên advice quá mạnh
25	Financial health score	Tổng hợp PFM	income stability, savings rate, debt ratio, bill punctuality	DS scoring/model	financial health level	Có product value, dễ demo	Dễ chủ quan nếu score không explainable
26	Savings opportunity detection	Tìm khoản có thể tiết kiệm	recurring subscriptions, category overspend, unused services	DS/rule + recommendation	saving opportunity	Hợp personal finance	Cần tránh tư vấn tài chính quá đà

27	Saving goal recommendation	User đặt mục tiêu tiết kiệm	income, spending surplus, recurring costs	Recommendation/rule	suggested monthly saving	Dễ hiểu, practical	AI/ML có thể nhẹ
28	Round-up saving simulator	Gợi ý tiết kiệm làm tròn giao dịch	transaction amounts, frequency, user goal	DS simulation	expected monthly saving	Rất dễ demo	Quá dễ nếu đứng riêng
29	Next best financial action	Personalized insight/action	persona, spending trend, bills, balance forecast	Ranking/recommendation	pay bill / save / reduce category	Bám personalization bank/ví	Dễ mơ hồ nếu action set không rõ
30	Financial persona segmentation	User profiling	spending mix, income pattern, merchant frequency, bill ratio	Clustering/DS	persona + explanation	Học Netflix/Spotify pattern map sang fintech	Khó evaluate persona đúng/sai

31	Lifestyle spending cluster	PFM segmentation	food, transport, shopping, entertainment proportions	Clustering	lifestyle segment	Dễ visualize, học DS segmentation	Có thể giống analytics bài tập
32	Bill-heavy user detection	Persona/action trigger	recurring bill ratio, fixed cost/income ratio	Rule/ML-lite	bill-heavy persona	Rất cụ thể, dễ dùng cho recommendation	Hơi đơn giản nếu đứng riêng
33	Cashflow unstable user detection	Persona/action trigger	income variance, balance volatility, late bills	ML/anomaly/DS	unstable cashflow persona	Hữu ích cho PFM/credit	Cần ethical framing
34	High subscription spender detection	Subscription insight	subscription count, total monthly recurring spend	DS/rule	subscription-heavy insight	Dễ demo, user hiểu ngay	AI/ML nhẹ
35	Impulse spending pattern detection	Behavioral insight	frequent small purchases, time-of-day, category spikes	DS/anomaly/clustering	impulse spend flag	Có behavioral analytics angle	Dễ chủ quan, cần cẩn thận

36	Merchant recommendation / offer targeting	Ví/app offer	merchant frequency, category preference, location giả lập	Recommendation/ranking	recommended offer/merchant	Bám MoMo/Techcombank personalization	Dễ thành marketing, cần metric
37	Promotion relevance ranking	Campaign targeting	user persona, past redemption, category spend	Ranking/ML	top offers	Có business value	Cần giả lập conversion labels
38	Bill payment recommendation	App nhắc thanh toán	detected bills, due date, balance forecast	Rule/ranking	pay now / remind later	Practical trong ngân hàng/ví	Không đủ ML nếu đứng riêng
39	Credit affordability pre-check from transactions	Pre-loan insight	income, spending, recurring bills, surplus cashflow	DS/rule + ML-lite	affordability estimate	Kết nối Open Banking + Credit	Nhạy cảm, không claim approval
40	Debt repayment capacity estimation	PFM/credit bridge	income, existing obligations, spending volatility	Regression/DS	estimated repayment capacity	Hợp lending fintech	Cần synthetic labels

41	Subscription cancellation suggestion	Cost optimization	inactive subscription, recurring payment, usage giả lập	DS/rule + ranking	cancellation candidate	User-facing practical	Cần tránh overreach nếu không có usage data
42	Fee detection and explanation	Bank fee transparency	transaction description, fee merchant/type, amount	NLP classification	fee category + explanation	Học explainable PFM	Phụ thuộc text patterns
43	Hidden fee / unusual charge alert	User charge monitoring	fee amount, historical fee pattern, merchant	Anomaly detection	unusual charge alert	Cụ thể, dễ demo	Dễ lẫn fraud/dispute
44	Refund tracking assistant	User theo dõi refund	original payment, refund transaction, merchant, date	Matching/DS	refund received / pending	Rất practical	Dữ liệu phải mô phỏng tốt

45	Split transaction / shared expense detection	PFM social spending	transfers after payment, notes, same amount patterns	DS/pattern mining	possible split expense	Hay, consumer app angle	Có thể xa banking core
46	Cash withdrawal pattern analysis	PFM behavior	ATM withdrawals, frequency, amount, location	DS/anomaly	cash usage insight	Học cashless transition behavior	Ít AI nếu đứng riêng
47	Spending after salary analysis	Behavioral finance	salary date, spend within 3/7 days, category spike	DS analytics	payday spending pattern	Cụ thể, dễ visualize	Không phải ML mạnh
48	Budget recommendation explanation	Explain PFM suggestion	budget, category trend, income, goals	LLM grounded explanation	explanation text	Hợp AI + UX	Nếu không có DS trước thì chỉ text
49	Personal finance chatbot over transactions	User hỏi về chi tiêu	transaction DB, categories, time filters	RAG/tool-calling + eval	answer with computed facts	Hợp AI app, có tool use	Dễ chatbot chung nếu thiếu evaluation

50	PFM insight quality evaluator	Evaluate generated insights	insight text, underlying facts, rules	LLM eval + rule checks	pass/fail insight quality	Học AI eval/guardrails	Nên làm add-on
51	Transaction search assistant	User tìm giao dịch	merchant, natural language query, time, amount	Semantic search/NLP	matching transactions	Hợp AI search, easy demo	Ít fintech-depth nếu đứng riêng
52	Spending query-to-SQL assistant	User hỏi bằng tiếng Việt	natural language, transaction schema	LLM-to-SQL + guardrails	SQL/result/chart	Hợp AI app + data	Cần guardrails tránh query sai
53	Financial goal progress tracker	Goal tracking	savings goal, monthly saving, balance trend	DS forecasting	goal on-track/off-track	User-facing, dễ hiểu	ML nhẹ
54	Scenario simulator: "Can I afford X?"	PFM planning	current balance, forecast, recurring bills, target purchase	DS simulation	affordability scenario	Practical, interesting	Không nên đưa financial advice thật

55	PFM data quality checker	Data pipeline monitoring	missing merchant, uncategorized rate, duplicate rate	DS analytics	data quality score	Học data engineering/quality	Không đủ AI/ML nếu đứng riêng
56	Open Banking data drift monitor	Long-term transaction model	category distribution, merchant distribution, income pattern	DS drift detection	drift alert	Hợp governance/data monitoring	Add-on hơn là project chính
57	User financial behavior change detection	Personalization update	persona over time, category shift, balance volatility	Change point/anomaly	behavior change alert	Học temporal analytics	Khó evaluate
58	Merchant category enrichment	Transaction enrichment	merchant name, MCC, location, description	NLP classification/embedding	enriched merchant category	Foundation tốt cho PFM	Dataset merchant khó

59	Cross-account transfer detection	Multi-account aggregation	same user accounts, equal opposite transactions	Matching/rule	internal transfer label	Cực hữu ích cho accurate spending	AI nhẹ, nhưng practical
60	Net worth / balance trend analysis	Account aggregation	balances across accounts, liabilities giả lập	DS analytics	net worth trend	User-facing, practical	Ít AI nếu đứng riêng

Package 1 — Open Banking Transaction Insight Prototype

Gồm:

- #1 Simulated Open Banking account aggregation
- #5 Transaction normalization
- #8 Transaction categorization
- #13 Unusual spending detection
- #16 Monthly spending summary

Package 2 — Personal Finance Persona & Recommendation Engine

Gồm:

- #8 Transaction categorization
- #10 Recurring bill detection
- #25 Financial health score

#30 Financial persona segmentation
#29 Next best financial action
#48 Budget recommendation explanation

Package 3 — Cashflow Early Warning Assistant

Gồm:

#20 Income detection
#21 Income stability scoring
#22 Cashflow forecasting
#23 Low balance early warning
#24 Cashflow stress detection

Package 4 — Subscription & Bill Intelligence

Gồm:

#10 Recurring bill detection
#11 Subscription detection
#12 Bill amount anomaly detection
#38 Bill payment recommendation
#41 Subscription cancellation suggestion

Package 5 — Transaction AI Assistant

Gồm:

#8 Transaction categorization
#49 Personal finance chatbot over transactions
#51 Transaction search assistant
#52 Spending query-to-SQL assistant
#50 PFM insight quality evaluator